5.4.1 Hermitian Positive Definite matrices ¶

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Hermitian Positive Definite (HPD) are a special class of matrices that are frequently encountered in practice.

Definition 5.4.1.1. Hermitian positive definite matrix. A matrix $A \in \mathbb{C}^{n \times n}$ is Hermitian positive definite (HPD) if and only if it is Hermitian ($A^H = A$) and for all nonzero vectors $x \in \mathbb{C}^n$ it is the case that $x^H A x > 0$. If in addition $A \in \mathbb{R}^{n \times n}$ then A is said to be symmetric positive definite (SPD).

If you feel uncomfortable with complex arithmetic, just replace the word "Hermitian" with "symmetric"" in this document and the Hermitian transpose operation, ^{H}, with the transpose operation, ^{T}.

Example 5.4.1.2. Consider the case where $n = 1$ so that A is a real scalar, α. Notice that then A is SPD if and only if $\alpha > 0$. This is because then for all nonzero $\chi \in \mathbb{R}$ it is the case that $\alpha \chi^2 > 0$.

Let's get some practice with reasoning about Hermitian positive definite matrices.

Homework 5.4.1.1. Let $B \in \mathbb{C}^{m \times n}$ have linearly independent columns.

ALWAYS/SOMETIMES/NEVER: $A = B^H B$ is HPD.

▾ Answer ▾ Solution

Let $x \in \mathbb{C}^m$ be a nonzero vector. Then $x^H B^H B x = (Bx)^H (Bx)$. Since B has linearly independent columns we know that $Bx \neq 0$. Hence $(Bx)^H Bx > 0$.

ALWAYS

Now prove it!

Homework 5.4.1.2. Let $A \in \mathbb{C}^{m \times m}$ be HPD.

ALWAYS/SOMETIMES/NEVER: The diagonal elements of A are real and positive.

▾ Hint ▾ Answer ▾ Solution

Let e_j be the jth unit basis vectors. Then $0 < e_j^H A e_j = \alpha_{j,j}$.

ALWAYS

Now prove it!

Consider the standard basis vector e_j.

Homework 5.4.1.3. Let $A \in \mathbb{C}^{m \times m}$ be HPD. Partition

$$A = \left(\begin{array}{c|c} \alpha_{11} & a_{21}^H \\ \hline a_{21} & A_{22} \end{array} \right).$$

ALWAYS/SOMETIMES/NEVER: A_{22} is HPD.

▸ Answer ▾ Solution

We need to show that $x_2^H A_{22} x_2 > 0$ for any nonzero $x_2 \in \mathbb{C}^{m-1}$.

Let $x_2 \in \mathbb{C}^{m-1}$ be a nonzero vector and choose $x = \begin{pmatrix} 0 \\ x_2 \end{pmatrix}$. Then

$$
\begin{array}{l}
0 \\
\quad < \quad \langle\ A \text{ is HPD } \rangle \\
x^H A x \\
\quad = \quad \langle\ \text{partition } \rangle \\
\left(\begin{array}{c} 0 \\ x_2 \end{array} \right)^H \left(\begin{array}{c|c} \alpha_{11} & a_{21}^H \\ \hline a_{21} & A_{22} \end{array} \right) \left(\begin{array}{c} 0 \\ x_2 \end{array} \right) \\
\quad = \quad \langle\ \text{multiply out } \rangle \\
x_2^H A_{22} x_2.
\end{array}
$$

We conclude that A_{22} is HPD.